



# HUST

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HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.





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# FUNDAMENTALS OF OPTIMIZATION

Linear Programming

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# CONTENT

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- Linear programs
- Geometric approach
- Simplex method
- Two-phase simplex method

# LINEAR PROGRAMS

- Standard form

$$f(x) = c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n \leq b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n \leq b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n \leq b_m$$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n \geq 0$$

# LINEAR PROGRAMS

- Standardize general linear programs
  - $f(x) \rightarrow \min \Leftrightarrow -f(x) \rightarrow \max$
  - $g(x) \geq b \Leftrightarrow -g(x) \leq -b$
  - $A = B \Leftrightarrow (A \leq B) \text{ and } (-A \leq -B)$
  - A variable  $x_j \in R$  can be represented by  $x_j = x_j^+ - x_j^-$  where  $x_j^+, x_j^- \geq 0$

# LINEAR PROGRAMS

- Example: Convert a general linear program forms into standard form

$$f(x_1, x_2) = 3x_1 + 2x_2 \rightarrow \min$$

$$2x_1 + x_2 \leq 7$$

$$x_1 + 2x_2 = 8$$

$$x_1 - x_2 \geq 2$$

$$x_1, x_2 \in \mathbb{R}, x_2 \geq 0$$

# LINEAR PROGRAMS

- Example: Convert a general linear program forms into standard form

- Substitution:  $x_1 = x_1^+ - x_1^-$

$$f(x_1^+, x_1^-, x_2) = -3x_1^+ + 3x_1^- - 2x_2 \rightarrow \max$$

$$2x_1^+ - 2x_1^- + x_2 \leq 7$$

$$x_1^+ - x_1^- + 2x_2 \leq 8$$

$$-x_1^+ + x_1^- - 2x_2 \leq -8$$

$$-x_1^+ + x_1^- + x_2 \leq -2$$

$$x_1^+, x_1^-, x_2 \in R, x_1^+, x_1^-, x_2 \geq 0$$



# GEOMETRIC APPROACH

- Constraints (inequalities) form a feasible region
- Optimal points will be one of the corners of the feasible region

# GEOMETRIC APPROACH

- Example

$$f(x_1, x_2) = 3x_1 + 2x_2 \rightarrow \max$$

$$2x_1 + x_2 \leq 7$$

$$x_1 + 2x_2 \leq 8$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

# GEOMETRIC APPROACH

- Example

$$f(x_1, x_2) = 3x_1 + 2x_2 \rightarrow \max$$

$$2x_1 + x_2 \leq 7$$

$$x_1 + 2x_2 \leq 8$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



# GEOMETRIC APPROACH

- Example

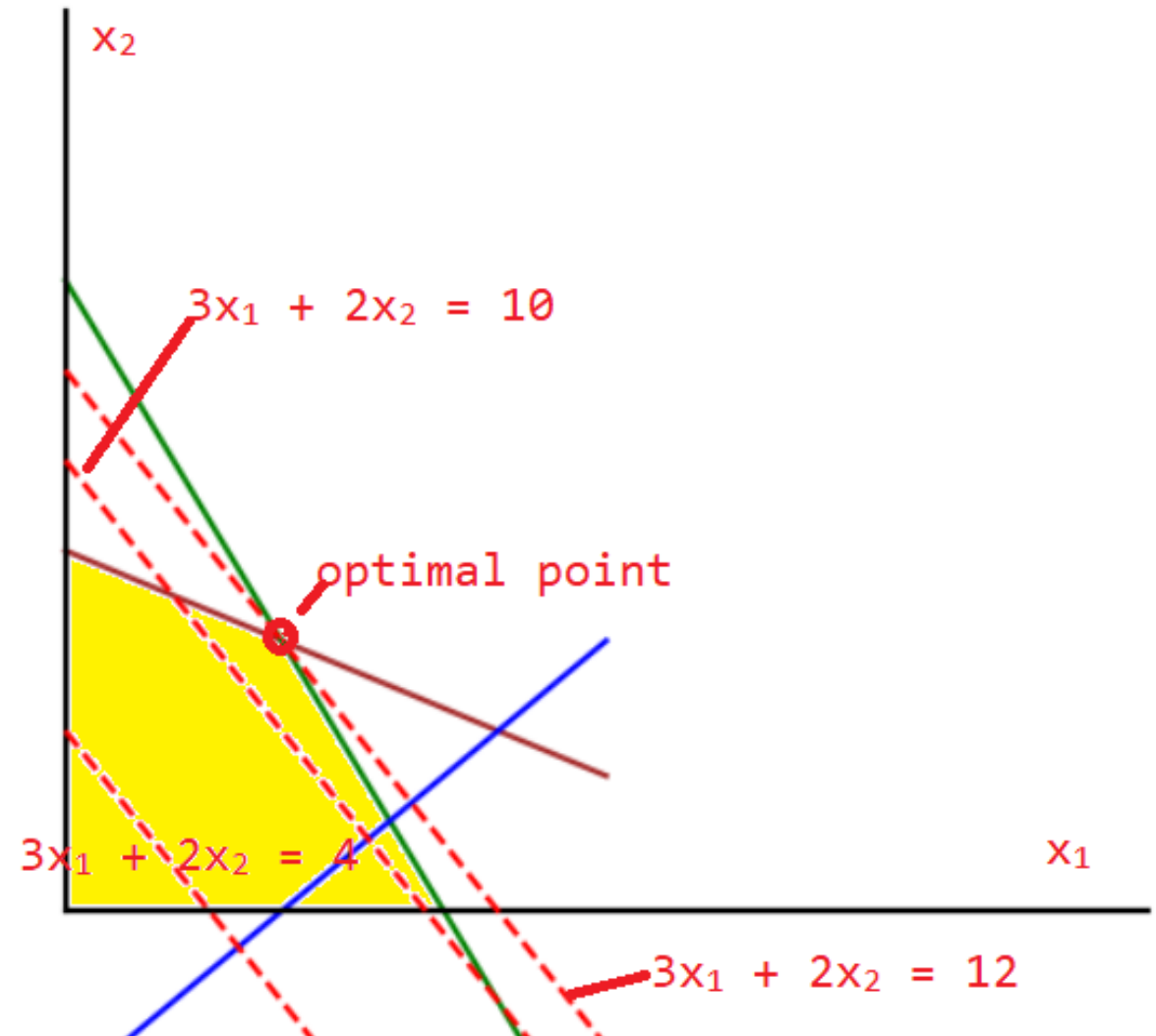
$$f(x_1, x_2) = 3x_1 + 2x_2 \rightarrow \max$$

$$2x_1 + x_2 \leq 7$$

$$x_1 + 2x_2 \leq 8$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0$$



# GEOMETRIC APPROACH

- Special cases
  - Problem does not have optimal solutions
  - Problem does not have feasible solutions

$$\begin{aligned}f(x_1, x_2) &= 3x_1 + 2x_2 \rightarrow \max \\-2x_1 - x_2 &\leq -7 \\x_1 - x_2 &\leq 2 \\x_1, x_2 &\in \mathbb{R}, x_1, x_2 \geq 0\end{aligned}$$

$$\begin{aligned}f(x_1, x_2) &= 3x_1 + 2x_2 \rightarrow \max \\2x_1 + x_2 &\leq 7 \\-4x_1 - 2x_2 &\leq -16 \\x_1, x_2 &\in \mathbb{R}, x_1, x_2 \geq 0\end{aligned}$$

- Exercise
  - A company must decide to make a plan to produce 2 products P1, P2.
    - The revenue received when selling 1 unit of P1 and P2 are respectively 5\$ and 7\$
    - The manufacturing cost when producing P1 and P2 are respectively 5\$ and 3\$
    - The storage cost in warehouses for 1 unit of P1 and P2 are respectively 2\$ and 3\$
  - Compute the production plan so that
    - Total manufacturing cost is less than or equal to 200\$
    - Total storage cost is less than or equal to 150\$
    - Total revenue is maximal

# BASIC FEASIBLE SOLUTION

- Standard form to standard equality form by adding slack variables  $y_1, y_2, \dots, y_m$

## Standard form

$$f(x) = c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n \leq b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n \leq b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n \leq b_m$$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n \geq 0$$



## Standard equality form

$$f(x) = c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n + y_1 = b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n + y_2 = b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n + y_m = b_m$$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m \geq 0$$

# BASIC FEASIBLE SOLUTION

- Consider a Linear Program (LP) under a standard equational form

Standard equational form

$$\begin{aligned} \max \quad & f(x) = c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \\ & a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n = b_1 \\ & a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n = b_2 \\ & \dots \\ & a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n = b_m \\ & x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n \geq 0 \end{aligned}$$



Standard equality form

$$\begin{aligned} f(x) &= c^T x \rightarrow \max \\ Ax &= b \\ x &\geq 0 \end{aligned}$$



# BASIC FEASIBLE SOLUTION

- Consider a Linear Program (LP) under a standard equational form
- Suppose  $\text{rank}(A) = m$
- Let  $B$  be the matrix of  $m$  linearly independent columns (indexed  $j_1, j_2, \dots, j_m$ ) of  $A$ :  $B = (A(j_1), A(j_2), \dots, A(j_m))$ 
  - Solution  $x$  is called a **basic solution** if :
    - $x_j = 0$  for  $j \in \{1, 2, \dots, n\} \setminus \{j_1, j_2, \dots, j_m\}$
    - Remain variables are found by solving this equation:

$$\begin{pmatrix} a_{1,j_1} & a_{1,j_2} & \dots & a_{1,j_m} \\ a_{2,j_1} & a_{2,j_2} & \dots & a_{2,j_m} \\ \dots & \dots & \dots & \dots \\ a_{m,j_1} & a_{m,j_2} & \dots & a_{m,j_m} \end{pmatrix} \begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \dots \\ x_{j_m} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{pmatrix}$$

$$\begin{aligned} f(x) &= c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max \\ a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n &= b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n &= b_2 \\ &\dots \\ a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n &= b_m \\ x_1, x_2, \dots, x_n &\in R, x_1, x_2, \dots, x_n \geq 0 \end{aligned}$$

# BASIC FEASIBLE SOLUTION

- Let  $B$  be the matrix of  $m$  linearly independent columns (indexed  $j_1, j_2, \dots, j_m$ ) of  $A$ :  $B = (A(j_1), A(j_2), \dots, A(j_m))$ 
  - Solution  $x$  is called a basic solution if :
    - $x_j = 0$  for  $j \in \{1, 2, \dots, n\} \setminus \{j_1, j_2, \dots, j_m\}$
    - Remain variables are found by solving this equation:

$$\begin{pmatrix} a_{1,j_1} & a_{1,j_2} & \dots & a_{1,j_m} \\ a_{2,j_1} & a_{2,j_2} & \dots & a_{2,j_m} \\ \dots & \dots & \dots & \dots \\ a_{m,j_1} & a_{m,j_2} & \dots & a_{m,j_m} \end{pmatrix} \begin{pmatrix} x_{j_1} \\ x_{j_2} \\ \dots \\ x_{j_m} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \dots \\ b_m \end{pmatrix}$$

- $B$  is called a basis
- $j_1, j_2, \dots, j_m$ : basic indices,  $j \in \{1, 2, \dots, n\} \setminus \{j_1, j_2, \dots, j_m\}$  is called non-basic index
- $x_{j_1}, x_{j_2}, \dots, x_{j_m}$ : basic variables and  $x_j$  ( $j \in \{1, 2, \dots, n\} \setminus \{j_1, j_2, \dots, j_m\}$ ) is called non-basic variable
- A **basic solution**  $x$  with  $x \geq 0$  is called a **basic feasible solution**

# SIMPLEX METHOD: Tabular form

- Consider a linear program under a standard form

$$\begin{aligned}
 z &= c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max \\
 a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n &\leq b_1 \\
 a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n &\leq b_2 \\
 &\dots \\
 a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n &\leq b_m \\
 b_1, b_2, \dots, b_m &\geq 0 \\
 x_1, x_2, \dots, x_n &\in R, x_1, x_2, \dots, x_n \geq 0
 \end{aligned}$$



$$\begin{aligned}
 z &= c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max \\
 a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n + y_1 &= b_1 \\
 a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n + y_2 &= b_2 \\
 &\dots \\
 a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n + y_m &= b_m \\
 b_1, b_2, \dots, b_m &\geq 0 \\
 x_1, x_2, \dots, x_n &\in R, x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m \geq 0
 \end{aligned}$$

|       | 1         | 2         | ... | $n$       | $n+1$ | $n+2$ | ... | $n+m$ |     |       |
|-------|-----------|-----------|-----|-----------|-------|-------|-----|-------|-----|-------|
| 0     | $x_1$     | $x_2$     | ... | $x_n$     | $y_1$ | $y_2$ | ... | $y_m$ | $z$ | RHS   |
| 1     | $a_{1,1}$ | $a_{1,2}$ | ... | $a_{1,n}$ | 1     | 0     | ... | 0     | 0   | $b_1$ |
| 2     | $a_{2,1}$ | $a_{2,2}$ | ... | $a_{2,n}$ | 0     | 1     | ... | 0     | 0   | $b_2$ |
| ...   | ...       | ...       | ... | ...       | ...   |       | ... | ...   | ..  | ...   |
| $m$   | $a_{m,1}$ | $a_{m,2}$ | ... | $a_{m,n}$ | 0     | 0     | ... | 1     | 0   | $b_m$ |
| $m+1$ | $-c_1$    | $-c_2$    | ... | $-c_n$    | 0     | 0     | ..  | 0     | 1   | 0     |

# SIMPLEX METHOD: Tabular form

|       | 1                | 2                | ... | $n$              | $n+1$              | $n+2$              | ... | $n+m$              | $n+m+1$              |               |
|-------|------------------|------------------|-----|------------------|--------------------|--------------------|-----|--------------------|----------------------|---------------|
| 0     | $x_1$            | $x_2$            | ... | $x_n$            | $x_{n+1}$          | $x_{n+2}$          | ... | $x_{n+m}$          | $z$                  | RHS           |
| 1     | $\alpha_{1,1}$   | $\alpha_{1,2}$   | ... | $\alpha_{1,n}$   | $\alpha_{1,n+1}$   | $\alpha_{1,n+2}$   | ... | $\alpha_{1,n+m}$   | $\alpha_{1,n+m+1}$   | $\beta_1$     |
| 2     | $\alpha_{2,1}$   | $\alpha_{2,2}$   | ... | $\alpha_{2,n}$   | $\alpha_{2,n+1}$   | $\alpha_{2,n+2}$   | ... | $\alpha_{2,n+m}$   | $\alpha_{2,n+m+1}$   | $\beta_2$     |
| ...   | ...              | ...              | ... | ...              | ...                | ...                | ... | ...                | ..                   | ...           |
| $m$   | $\alpha_{m,1}$   | $\alpha_{m,2}$   | ... | $\alpha_{m,n}$   | $\alpha_{m,n+1}$   | $\alpha_{m,n+2}$   | ... | $\alpha_{m,n+m}$   | $\alpha_{m,n+m+1}$   | $\beta_m$     |
| $m+1$ | $\alpha_{m+1,1}$ | $\alpha_{m+1,2}$ | ... | $\alpha_{m+1,n}$ | $\alpha_{m+1,n+1}$ | $\alpha_{m+1,n+2}$ | ... | $\alpha_{m+1,n+m}$ | $\alpha_{m+1,n+m+1}$ | $\beta_{m+1}$ |

- $J = \{1, 2, \dots, n, n+1, \dots, n+m\}$
- Maintain linear constraints on each row  $k$  ( $k = 1, 2, \dots, m+1$ ):

$$\alpha_{k,1}x_1 + \alpha_{k,2}x_2 + \dots + \alpha_{k,n}x_n + \alpha_{k,n+1}x_{n+1} + \dots + \alpha_{k,n+m}x_{n+m} + \alpha_{k,n+m+1}z = \beta_k \quad (*)$$

- Let  $R_k$  be a vector containing elements on row  $k$  of the table ( $k = 1, 2, \dots, m+1$ )
- Perform linear transformation below, constraint (\*) is still satisfied:
  - Replace  $R_k = R_k + \delta^* R_i$  ( $k, i = 1, 2, \dots, m+1$ ), with some constant  $\delta$

# SIMPLEX METHOD: Tabular form

- Optimality

|       | 1     | 2     | ... | $m$   | $m+1$              | $m+2$              | ... | $n+m$              | $n+m+1$ |               |
|-------|-------|-------|-----|-------|--------------------|--------------------|-----|--------------------|---------|---------------|
| 0     | $x_1$ | $x_2$ | ... | $x_m$ | $x_{m+1}$          | $x_{m+2}$          | ... | $x_{n+m}$          | $z$     | RHS           |
| 1     | 1     | 0     | ... | 0     | $\alpha_{1,m+1}$   | $\alpha_{1,m+2}$   | ... | $\alpha_{1,n+m}$   | 0       | $\beta_1$     |
| 2     | 0     | 1     | ... | 0     | $\alpha_{2,m+1}$   | $\alpha_{2,m+2}$   | ... | $\alpha_{2,n+m}$   | 0       | $\beta_2$     |
| ...   | ...   | ...   | ... | ...   | ...                | ...                | ... | ...                | ..      | ...           |
| $m$   | 0     | 0     | ... | 1     | $\alpha_{m,m+1}$   | $\alpha_{m,m+2}$   | ... | $\alpha_{m,n+m}$   | 0       | $\beta_m$     |
| $m+1$ | 0     | 0     | ... | 0     | $\alpha_{m+1,m+1}$ | $\alpha_{m+1,m+2}$ | ... | $\alpha_{m+1,n+m}$ | 1       | $\beta_{m+1}$ |

- With  $\beta_1, \beta_2, \dots, \beta_m \geq 0$ ,  $\exists J_B = \{j_1, j_2, \dots, j_m\}$  such that  $\alpha_{m+1,j} = 0, \forall j \in J_B, \alpha_{m+1,j} \geq 0 \forall j \in J \setminus J_B$ , columns  $j_1, j_2, \dots, j_m$  forms a unit matrix
- Without loss of generality, suppose that  $J_B = \{1, 2, \dots, n\}$ , coefficients  $\alpha_{m+1,m+1}, \alpha_{m+1,m+2}, \dots, \alpha_{m+1,n+m} \geq 0$ , columns 1, ..., m forms a unit matrix:  $\alpha_{1,1}, \alpha_{2,2}, \dots, \alpha_{m,m} = 1$
- Constraint (\*) is still satisfied. We have  $\alpha_{m+1,m+1}x_{m+1} + \alpha_{m+1,m+2}x_{m+2} + \dots, \alpha_{m+1,n+m}x_{n+m} + z = \beta_{m+1}$
- $z = \beta_{m+1} - (\alpha_{m+1,m+1}x_{m+1} + \alpha_{m+1,m+2}x_{m+2} + \dots, \alpha_{m+1,n+m}x_{n+m}) \leq \beta_{m+1}$  (because  $\alpha_{m+1,m+1}, \alpha_{m+1,m+2}, \dots, \alpha_{m+1,n+m} \geq 0$  and  $x_{m+1}, \dots, x_{n+m} \geq 0$ ).

# SIMPLEX METHOD: Tabular form

- Optimality

|       | 1     | 2     | ... | $m$   | $m+1$              | $m+2$              | ... | $n+m$              | $n+m+1$ |               |
|-------|-------|-------|-----|-------|--------------------|--------------------|-----|--------------------|---------|---------------|
| 0     | $x_1$ | $x_2$ | ... | $x_m$ | $x_{m+1}$          | $x_{m+2}$          | ... | $x_{n+m}$          | $z$     | RHS           |
| 1     | 1     | 0     | ... | 0     | $\alpha_{1,m+1}$   | $\alpha_{1,m+2}$   | ... | $\alpha_{1,n+m}$   | 0       | $\beta_1$     |
| 2     | 0     | 1     | ... | 0     | $\alpha_{2,m+1}$   | $\alpha_{2,m+2}$   | ... | $\alpha_{2,n+m}$   | 0       | $\beta_2$     |
| ...   | ...   | ...   | ... | ...   | ...                | ...                | ... | ...                | ..      | ...           |
| $m$   | 0     | 0     | ... | 1     | $\alpha_{m,m+1}$   | $\alpha_{m,m+2}$   | ... | $\alpha_{m,n+m}$   | 0       | $\beta_m$     |
| $m+1$ | 0     | 0     | ... | 0     | $\alpha_{m+1,m+1}$ | $\alpha_{m+1,m+2}$ | ... | $\alpha_{m+1,n+m}$ | 1       | $\beta_{m+1}$ |

- Moreover, there exists a solution (nonnegative values for variables  $x_1, x_2, \dots, x_{n+m}$ ) described below:

- $x_1 = \beta_1, x_2 = \beta_2, \dots, x_m = \beta_m$
- $x_{m+1} = x_{m+2} = \dots = x_{n+m} = 0$

Satisfying given constraints. Also, the objective value at this solution is equal to the upper bound  $\beta_{m+1}$ . It means that this solution is an optimal solution to the given problem.

# SIMPLEX METHOD: Tabular form

- Simplex step

|       | 1     | 2     | ... | $m$   | $m+1$              |     | $i$              | ... | $n+m$              |     |               |       |
|-------|-------|-------|-----|-------|--------------------|-----|------------------|-----|--------------------|-----|---------------|-------|
| 0     | $x_1$ | $x_2$ | ... | $x_m$ | $x_{m+1}$          | ... | $x_i$            | ... | $x_{n+m}$          | $z$ | RHS           | $E$   |
| 1     | 1     | 0     | ... | 0     | $\alpha_{1,m+1}$   | ... | $\alpha_{1,i}$   | ... | $\alpha_{1,n+m}$   | 0   | $\beta_1$     | $E_1$ |
| 2     | ...   | ...   | ... | ...   | ...                | ... | ...              | ... | ...                | ... | ...           | ...   |
| ...   | 0     | 1     | ... | 0     | $\alpha_{k,m+1}$   | ... | $\alpha_{k,i}$   | ... | $\alpha_{k,n+m}$   | 0   | $\beta_k$     | $E_k$ |
| $m$   | 0     | 0     | ... | 1     | $\alpha_{m,m+1}$   | ... | $\alpha_{m,i}$   | ... | $\alpha_{m,n+m}$   | 0   | $\beta_m$     | $E_m$ |
| $m+1$ | 0     | 0     | ... | 0     | $\alpha_{m+1,m+1}$ | ... | $\alpha_{m+1,i}$ | ... | $\alpha_{m+1,n+m}$ | 1   | $\beta_{m+1}$ |       |

- Select column  $i$  such that the element on row  $m+1$  (which is  $\alpha_{m+1,i}$ ) is negative minimal
- Compute evaluations (column E):  $E_j = +\infty$ , if  $\alpha_{j,i} \leq 0$ , and  $E_j = \frac{\beta_i}{\alpha_{j,i}}$ , if  $\alpha_{j,i} > 0$ ,  $j = 1, 2, \dots, m$
- Select the row  $k$  such that  $E_k$  is minimal: if  $E_k = +\infty$ , then the problem is **unbounded**, otherwise
  - Update:
    - Row  $R_k = R_k / \alpha_{k,i}$
    - Row  $R_j = R_j - \alpha_{j,i} * R_k$ ,  $j = \{1, 2, \dots, m+1\} \setminus \{k\}$

# SIMPLEX METHOD: Tabular form

- Example

$$\begin{aligned} z &= 3x_1 + 2x_2 \rightarrow \max \\ 2x_1 + x_2 &\leq 7 \\ x_1 + 2x_2 &\leq 8 \\ x_1 - x_2 &\leq 2 \\ x_1, x_2 &\in R, x_1, x_2 \geq 0 \end{aligned}$$



$$\begin{aligned} z &= 3x_1 + 2x_2 \rightarrow \max \\ 2x_1 + x_2 + x_3 &= 7 \\ x_1 + 2x_2 + x_4 &= 8 \\ x_1 - x_2 + x_5 &= 2 \\ x_1, x_2, x_3, x_4, x_5 &\in R, x_1, x_2, x_3, x_4, x_5 \geq 0 \end{aligned}$$

| $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|-------|-------|-------|-------|-------|-----|-----|---|
| 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |



# SIMPLEX METHOD: Tabular form

- Example

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |

# SIMPLEX METHOD: Tabular form

- Example

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |

- Select column 1: the corresponding element at the last row is minimal



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |

- Select column 1: the corresponding element at the last row is minimal
- Compute evaluation (column E)



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E   |
|---|-------|-------|-------|-------|-------|-----|-----|-----|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   | 7/2 |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   | 8/1 |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   | 2/1 |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |     |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |

- Select column 1: the corresponding element at the last row is minimal
- Compute evaluation (column E)
- Select row R3: evaluation is minimal
- Update  $R3 = R3/1$



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E   |
|---|-------|-------|-------|-------|-------|-----|-----|-----|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   | 7/2 |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   | 8/1 |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   | 2/1 |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |     |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 2     | 1     | 1     | 0     | 0     | 0   | 7   |   |
| 2 | 1     | 2     | 0     | 1     | 0     | 0   | 8   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | -3    | -2    | 0     | 0     | 0     | 1   | 0   |   |

- Select column 1: the corresponding element at the last row is minimal
- Compute evaluation (column E)
- Select row R3: evaluation is minimal
- Update  $R3 = R3/1$
- $R1 = R1 - 2R3$ ;  $R2 = R2 - R3$ ;  $R4 = R4 + 3R3$ ;

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

- Select column 2: the corresponding element at the last row is minimal



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

- Select column 2: the corresponding element at the last row is minimal
- Compute evaluations: column E



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E         |
|---|-------|-------|-------|-------|-------|-----|-----|-----------|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   | 3/3       |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   | 6/3       |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   | $+\infty$ |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |           |



# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

- Select column 2: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row R1: minimum evaluation



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E         |
|---|-------|-------|-------|-------|-------|-----|-----|-----------|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   | 3/3       |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   | 6/3       |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   | $+\infty$ |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |           |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

- Select column 2: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row R1: minimum evaluation
- Update  $R1 = R1/3$



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 3     | 1     | 0     | -2    | 0   | 3   |   |
| 2 | 0     | 3     | 0     | 1     | -1    | 0   | 6   |   |
| 3 | 1     | -1    | 0     | 0     | 1     | 0   | 2   |   |
| 4 | 0     | -5    | 0     | 0     | 3     | 1   | 6   |   |

- Select column 2: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row R1: minimum evaluation
- Update  $R1 = R1/3$
- $R2 = R2 - 3R1$ ;  $R3 = R3 + R1$ ;  $R4 = R4 + 5R1$

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$  | $z$ | RHS | E |
|---|-------|-------|-------|-------|--------|-----|-----|---|
| 1 | 0     | 1     | $1/3$ | 0     | $-2/3$ | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1      | 0   | 3   |   |
| 3 | 1     | 0     | $1/3$ | 0     | $1/3$  | 0   | 3   |   |
| 4 | 0     | 0     | $5/3$ | 0     | $-1/3$ | 1   | 11  |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

- Select column 5: the corresponding element at the last row is minimal



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

- Select column 5: the corresponding element at the last row is minimal
- Compute evaluations: column E



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E         |
|---|-------|-------|-------|-------|-------|-----|-----|-----------|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   | $+\infty$ |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   | 3         |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   | 9         |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |           |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

- Select column 5: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row 2: minimum evaluation



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E         |
|---|-------|-------|-------|-------|-------|-----|-----|-----------|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   | $+\infty$ |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   | 3         |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   | 9         |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |           |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

- Select column 5: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row 2: minimum evaluation
- Update:  $R2 = R2/1$



|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |



# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

- Select column 5: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row 2: minimum evaluation
- Update:  $R2 = R2/1$
- $R1 = R1 + (2/3)R2$ ;  $R3 = R3 - (1/3)R2$ ;  $R4 = R4 + (1/3)R2$

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | -1/3  | 2/3   | 0     | 0   | 3   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 2/3   | -1/3  | 0     | 0   | 2   |   |
| 4 | 0     | 0     | 4/3   | 1/3   | 0     | 1   | 12  |   |

# SIMPLEX METHOD: Tabular form

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | 1/3   | 0     | -2/3  | 0   | 1   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 1/3   | 0     | 1/3   | 0   | 3   |   |
| 4 | 0     | 0     | 5/3   | 0     | -1/3  | 1   | 11  |   |

- Select column 5: the corresponding element at the last row is minimal
- Compute evaluations: column E
- Select row 2: minimum evaluation
- Update:  $R2 = R2/1$
- $R1 = R1 + (2/3)R2$ ;  $R3 = R3 - (1/3)R2$ ;  $R4 = R4 + (1/3)R2$

|   | $x_1$ | $x_2$ | $x_3$ | $x_4$ | $x_5$ | $z$ | RHS | E |
|---|-------|-------|-------|-------|-------|-----|-----|---|
| 1 | 0     | 1     | -1/3  | 2/3   | 0     | 0   | 3   |   |
| 2 | 0     | 0     | -1    | 1     | 1     | 0   | 3   |   |
| 3 | 1     | 0     | 2/3   | -1/3  | 0     | 0   | 2   |   |
| 4 | 0     | 0     | 4/3   | 1/3   | 0     | 1   | 12  |   |

- Phương án tối ưu của bài toán là  $x_1 = 2$  và  $x_2 = 3$ ,  $x_3 = 0$ ,  $x_4 = 0$ ,  $x_5 = 3$ .
- Giá trị hàm mục tiêu tối ưu bằng 12

# TWO-PHASE SIMPLEX METHOD

- Consider a linear program under a standard equational form

$$(LP) \quad z = c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n = b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n = b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n = b_m$$

Coefficients  $b_1, b_2, \dots, b_m \geq 0$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n \geq 0$$

# TWO-PHASE SIMPLEX METHOD

- Consider a linear program under a standard equational form

$$(LP) \quad z = c_1x_1 + c_2x_2 + \dots + c_nx_n \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n = b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n = b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n = b_m$$

$$\text{Coefficients } b_1, b_2, \dots, b_m \geq 0$$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n \geq 0$$

Introduce an auxiliary linear program (ALP) with  $m$  artificial variables  $y_1, y_2, \dots, y_m$

$$(ALP) \quad g = -y_1 - y_2 - \dots - y_m \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n + y_1 = b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n + y_2 = b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n + y_m = b_m$$

$$b_1, b_2, \dots, b_m \geq 0$$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m \geq 0$$

# TWO-PHASE SIMPLEX METHOD

$$(ALP) \quad g = -y_1 - y_2 - \dots - y_m \rightarrow \max$$

$$a_{1,1}x_1 + a_{1,2}x_2 + \dots + a_{1,n}x_n + y_1 = b_1$$

$$a_{2,1}x_1 + a_{2,2}x_2 + \dots + a_{2,n}x_n + y_2 = b_2$$

...

$$a_{m,1}x_1 + a_{m,2}x_2 + \dots + a_{m,n}x_n + y_m = b_m$$

$$b_1, b_2, \dots, b_m \geq 0$$

$$x_1, x_2, \dots, x_n \in R, x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_m \geq 0$$

- Solve the (ALP) by the Simplex Method under the Tabular form: Basis is the column vectors corresponding to artificial variables  $\rightarrow$  obtain an optimal solution  $(x^*, y^*)$  and basic indices set is  $J_B^*$
- Proposition: The original (LP) problem has feasible solutions iff the corresponding (ALP) has an optimal solution  $(x^*, y^*)$  in which  $y^* = 0$  (**proof ?**)
- If  $y^* \neq 0$ , then the original (LP) problem does not have feasible solutions
- We consider the case that  $y^* = 0$

# TWO-PHASE SIMPLEX METHOD

|       | 1                | 2                | ... | $n$              | $n+1$              | $n+2$              | ... | $n+m$              | $n+m+1$ |               |
|-------|------------------|------------------|-----|------------------|--------------------|--------------------|-----|--------------------|---------|---------------|
| 0     | $x_1$            | $x_2$            | ... | $x_n$            | $y_1$              | $y_2$              | ... | $y_m$              | $z$     | RHS           |
| 1     | $\alpha_{1,1}$   | $\alpha_{1,2}$   | ... | $\alpha_{1,n}$   | $\alpha_{1,n+1}$   | $\alpha_{1,n+2}$   | ... | $\alpha_{1,n+m}$   | 0       | $\beta_1$     |
| 2     | $\alpha_{2,1}$   | $\alpha_{2,2}$   | ... | $\alpha_{2,n}$   | $\alpha_{2,n+1}$   | $\alpha_{2,n+2}$   | ... | $\alpha_{2,n+m}$   | 0       | $\beta_2$     |
| ...   | ...              | ...              | ... | ...              | ...                | ...                | ... | ...                | ..      | ...           |
| $m$   | $\alpha_{m,1}$   | $\alpha_{m,2}$   | ... | $\alpha_{m,n}$   | $\alpha_{m,n+1}$   | $\alpha_{m,n+2}$   | ... | $\alpha_{m,n+m}$   | 0       | $\beta_m$     |
| $m+1$ | $\alpha_{m+1,1}$ | $\alpha_{m+1,2}$ | ... | $\alpha_{m+1,n}$ | $\alpha_{m+1,n+1}$ | $\alpha_{m+1,n+2}$ | ... | $\alpha_{m+1,n+m}$ | 1       | $\beta_{m+1}$ |

- Case 1:  $J_B^*$  does not contain indices of artificial variables
  - Move to the second phase, solve the original (LP) problem
    - Remove columns corresponding to artificial variables:  $n+1, \dots, n+m$
    - Recompute elements on row  $m+1$  (based on the original objective function)

# TWO-PHASE SIMPLEX METHOD

|       | 1                | 2                | ... | $n$              | $n+1$              | $n+2$              | ... | $n+m$              | $n+m+1$ |               |
|-------|------------------|------------------|-----|------------------|--------------------|--------------------|-----|--------------------|---------|---------------|
| 0     | $x_1$            | $x_2$            | ... | $x_n$            | $y_1$              | $y_2$              | ... | $y_m$              | $z$     | RHS           |
| 1     | $\alpha_{1,1}$   | $\alpha_{1,2}$   | ... | $\alpha_{1,n}$   | $\alpha_{1,n+1}$   | $\alpha_{1,n+2}$   | ... | $\alpha_{1,n+m}$   | 0       | $\beta_1$     |
| 2     | $\alpha_{2,1}$   | $\alpha_{2,2}$   | ... | $\alpha_{2,n}$   | $\alpha_{2,n+1}$   | $\alpha_{2,n+2}$   | ... | $\alpha_{2,n+m}$   | 0       | $\beta_2$     |
| ...   | ...              | ...              | ... | ...              | ...                | ...                | ... | ...                | ..      | ...           |
| $m$   | $\alpha_{m,1}$   | $\alpha_{m,2}$   | ... | $\alpha_{m,n}$   | $\alpha_{m,n+1}$   | $\alpha_{m,n+2}$   | ... | $\alpha_{m,n+m}$   | 0       | $\beta_m$     |
| $m+1$ | $\alpha_{m+1,1}$ | $\alpha_{m+1,2}$ | ... | $\alpha_{m+1,n}$ | $\alpha_{m+1,n+1}$ | $\alpha_{m+1,n+2}$ | ... | $\alpha_{m+1,n+m}$ | 1       | $\beta_{m+1}$ |

- Case 2:  $J_B^*$  contains some indices of artificial variables
  - Suppose  $J_B^*$  contains index  $n+j$  of the artificial variable ( $y_j$ ), perform the linear transformation to remove index  $n+j$  from  $J_B^*$  as follows:
    - Consider row  $k$  such that the element in row  $k$  and column  $n+j$  is 1 (column vector corresponding to column  $n+j$  is a unit vector)
    - Case 2.1: If all elements on row  $k$ , from column 1 to column  $n$  are equal to 0 ( $\alpha_{k,1} = \dots = \alpha_{k,n} = 0$ ): it means, the constraint of row  $k$  is linear dependent on other constraints  $\rightarrow$  we can remove this row  $k$  and column  $n+j$  from the table

# TWO-PHASE SIMPLEX METHOD

|       | 1                | 2                | ... | $n$              | $n+1$              | $n+2$              | ... | $n+m$              | $n+m+1$ |               |
|-------|------------------|------------------|-----|------------------|--------------------|--------------------|-----|--------------------|---------|---------------|
| 0     | $x_1$            | $x_2$            | ... | $x_n$            | $y_1$              | $y_2$              | ... | $y_m$              | $z$     | RHS           |
| 1     | $\alpha_{1,1}$   | $\alpha_{1,2}$   | ... | $\alpha_{1,n}$   | $\alpha_{1,n+1}$   | $\alpha_{1,n+2}$   | ... | $\alpha_{1,n+m}$   | 0       | $\beta_1$     |
| 2     | $\alpha_{2,1}$   | $\alpha_{2,2}$   | ... | $\alpha_{2,n}$   | $\alpha_{2,n+1}$   | $\alpha_{2,n+2}$   | ... | $\alpha_{2,n+m}$   | 0       | $\beta_2$     |
| ...   | ...              | ...              | ... | ...              | ...                | ...                | ... | ...                | ..      | ...           |
| $m$   | $\alpha_{m,1}$   | $\alpha_{m,2}$   | ... | $\alpha_{m,n}$   | $\alpha_{m,n+1}$   | $\alpha_{m,n+2}$   | ... | $\alpha_{m,n+m}$   | 0       | $\beta_m$     |
| $m+1$ | $\alpha_{m+1,1}$ | $\alpha_{m+1,2}$ | ... | $\alpha_{m+1,n}$ | $\alpha_{m+1,n+1}$ | $\alpha_{m+1,n+2}$ | ... | $\alpha_{m+1,n+m}$ | 1       | $\beta_{m+1}$ |

- Case 2:  $J_B^*$  contains some indices of artificial variables
  - Case 2.2: There exists a column  $i$  such that  $\alpha_{k,i} \neq 0$
  - In this optimal table, all artificial variables are equal to 0, so  $\beta_k$  is equal to 0
  - Perform the rotation with the pivot  $\alpha_{k,i}$ . With this rotation, column RHS is unchanged due to the fact that  $\beta_k$  is equal to 0. Hence  $\beta_{m+1}$  is always 0. It means that the new table corresponds to another optimal solution in which one artificial variable is replaced by an original variable  $x_i$
  - The above procedure is repeated until all artificial variables are removed from the basic
  - We now process the computation as the case 1 (above)



# TWO-PHASE SIMPLEX METHOD

- Example 1 (exercise in class)

$$(LP) \quad z = x_1 + 2x_2 - x_3 + x_4 \rightarrow \max$$

$$x_1 + x_2 - x_3 - x_4 = 4$$

$$x_1 + x_3 + x_4 = 7$$

$$2x_1 + x_2 = 2$$

$$x_1, x_2, x_3, x_4 \in R, x_1, x_2, x_3, x_4 \geq 0$$

# TWO-PHASE SIMPLEX METHOD

- Example 2 (exercise in class)

$$(LP) \quad z = x_1 + 2x_2 - x_3 + x_4 \rightarrow \max$$

$$x_1 + x_2 - x_3 - x_4 = 4$$

$$x_1 + x_3 + x_4 = 7$$

$$x_1 - x_2 - x_3 = 2$$

$$x_1, x_2, x_3, x_4 \in R, x_1, x_2, x_3, x_4 \geq 0$$

# TWO-PHASE SIMPLEX METHOD

- Example 3 (exercise in class)

$$\begin{aligned} \text{(LP)} \quad z &= 40x_1 + 10x_2 + 7x_5 + 14x_6 \rightarrow \max \\ x_1 - x_2 &\quad + 2x_5 &= 0 \\ -2x_1 + x_2 &\quad - 2x_5 &= 0 \\ x_1 &\quad + x_3 + x_5 - x_6 &= 3 \\ x_2 + x_3 + x_4 + 2x_5 + x_6 &= 4 \\ x_1, x_2, x_3, x_4, x_5, x_6 &\in R, x_1, x_2, x_3, x_4, x_5, x_6 \geq 0 \end{aligned}$$

A large graphic on the left side of the slide. It features a dark blue background with a circular pattern of red dots of varying sizes, creating a sense of depth and movement. The word "HUST" is centered within this pattern in a bold, white, sans-serif font.

# HUST

# THANK YOU !